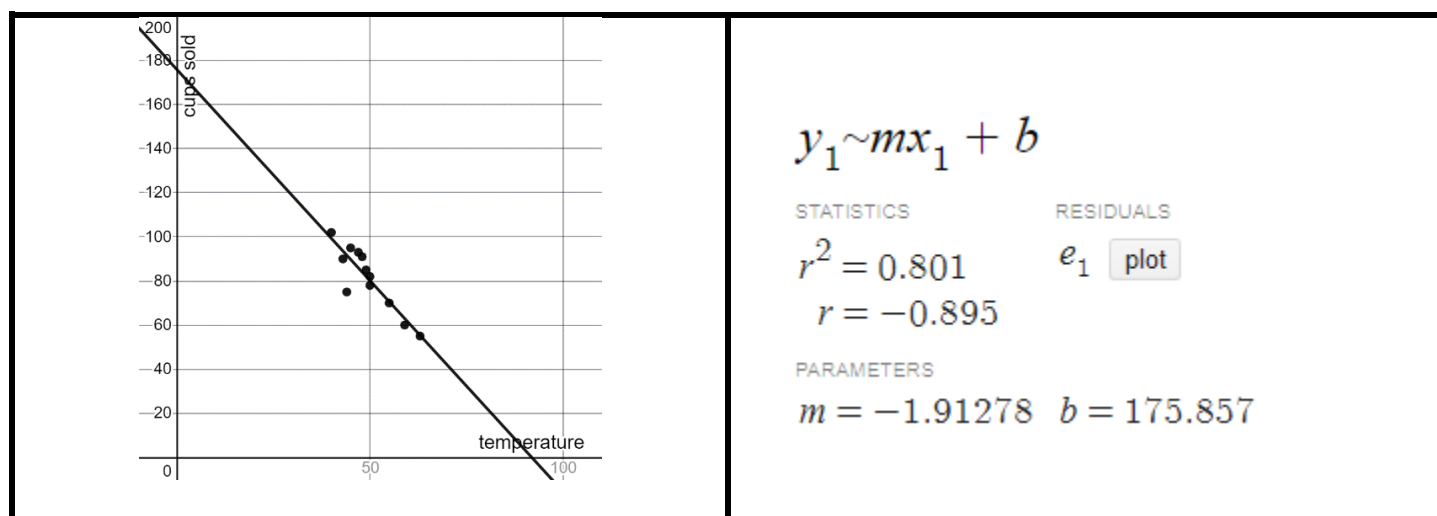


Use this scenario for questions 1 through 6. Connie sells hot chocolate at a stand in a park. She records the number of cups of hot chocolate she sells, as well as the average daily low temperature. The table shown indicates the data for 12 days in January.

Hot Chocolate Sales

Average Temperature (°F)	59	63	55	50	49	40	45	43	47	44	50	48
Cups of Hot Chocolate Sold	60	55	70	82	85	102	95	90	93	75	78	91

Connie used technology to generate a scatter plot and a line of best fit, as shown.



- Write the equation for the line of best fit using the values generated with technology.
 $y = -1.91278x + 175.857$
- Interpret the slope of the line of best fit in the context.
There are 1.91278 less cups of hot chocolate sold per degree increase of temperature.
- Describe the strength of the association between temperature and the number of cups of hot chocolate Connie sells.
Since the r (correlation coefficient) is near -1 , this shows there is a strong association between temperature and the number of cups of hot chocolate sold.

4. Describe the direction of the linear association between temperature and the number of cups of hot chocolate Connie sells.

As the temperature increases, the number of cups of hot chocolate decreases – resulting in a negative linear association.

5. Suppose Connie knows that tomorrow's temperature is predicted to be 40°F. Based on the line of best fit, how many cups of hot chocolate should she anticipate selling?

$$y = -1.91278(40) + 175.857$$

$$y = 99.3458$$

99 cups

6. A few weeks ago, Connie sold 120 cups of hot chocolate. Based on the line of best fit, what was the average temperature that day?

$$120 = -1.91278x + 175.857$$

$$-120 = -1.91278x - 175.857$$

$$\frac{-55.857}{-1.91278} = \frac{-1.91278x}{-1.91278}$$

$$x = 29.2$$

29°F

A scatter plot created with technology generated the following values, as shown. Use the values for questions 7 through 10.

$$y_1 \sim mx_1 + b$$

STATISTICS

$$r^2 = 0.8913$$

$$r = 0.9441$$

PARAMETERS

$$m = 0.860687$$

RESIDUALS

$$e_1 \text{ plot}$$

$$b = 0.645038$$

7. Describe the strength of the linear association for this data.

Since the r (correlation coefficient) is close to 1, there is a strong association.

8. Describe the direction of the linear association of this data.

Because the slope (or the correlation coefficient) is positive, the direction of the line of best fit is positive – increasing from left to right.

9. Use this data to write the equation of the line of best fit in slope-intercept form.

$$y = 0.860687x + 0.645038$$

10. Use the line of best fit to predict a value for y if the value of x is 18.

$$y = 0.860687(18) + 0.645038$$

$$y = 16.137404$$